

Aritra Das

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Google Scholar: scholar.google.com/citations?user=WkuhmCUAAAAJ

PROFESSIONAL SUMMARY

Computer Science researcher and AI developer with an MS in Computer Science and Engineering and 11 peer-reviewed publications, including six Q1 journal papers. Research experience spans machine learning, deep learning, natural language processing, computer vision, self-supervised learning, and large language model applications, with work involving multilingual text analysis, benchmark dataset creation, and malware classification. Recipient of the Best Paper Award at All 2025. Skilled in Python, PyTorch, TensorFlow, Transformer-based models, LangChain, and RAG pipelines.

EDUCATION

MS in Computer Science and Engineering, East West University, Dhaka, Bangladesh

January 2025 – December 2025 | CGPA: 3.91 / 4.00

B.Sc. in Computer Science and Engineering, East West University, Dhaka, Bangladesh

January 2020 – December 2024 | CGPA: 3.76 / 4.00

Higher Secondary Certificate (HSC), Dhaka College, Dhaka, Bangladesh

June 2017 – July 2019 | GPA: 4.58 / 5.00

Secondary School Certificate (SSC), Ideal School and College, Motijheel, Dhaka, Bangladesh

January 2007 – May 2017 | GPA: 5.00 / 5.00

PROFESSIONAL EXPERIENCE

AI Developer, SM Technology, Dhaka, Bangladesh

April 2026 – Present

- Develop and deploy AI solutions, including LLM-based applications and RAG pipelines, from prototyping through production.

Teaching Assistant, East West University, Dhaka, Bangladesh

January 2024 – December 2025

- Assisted with laboratory instruction, assignment evaluation, and student consultation for Computer Science and Engineering courses.

Machine Learning Engineer, ACME AI, Dhaka, Bangladesh

June 2024 – September 2024

- Built and evaluated machine learning models, covering data annotation, model training, and performance benchmarking.

PUBLICATIONS: JOURNAL ARTICLES

- A. Das, M. R. A. Rashid, M. R. Hasan, K. Shams, and R. U. Islam, "BDFlower: Growth stage flower image dataset for precision agriculture and floriculture," *Data in Brief*, vol. 66, p. 112745, 2026, doi: 10.1016/j.dib.2026.112745. (Q1)
- A. Das, K. Shams, M. R. A. Rashid, R. U. Islam, S. H. Ripon, and A. W. Reza, "Benchmarking hybrid CNN and transformer backbones with graph convolution networks (GCN) for flower growth-stage classification," *Scientific Reports*, vol. 16, art. no. 56866, June 2026, doi: 10.1038/s41598-026-56866-y. (Q1)
- M. R. A. Rashid, A. Das, K. F. Hasan, M. R. Hasan, M. Sultana, M. Hasan, R. U. Islam, R. A. Tuhin, and M. S. H. Khan, "BERT-KAN: Enhancing bilingual sentiment analysis in Bangladeshi e-commerce through fine-tuned large language models," *Natural Language Processing Journal*, vol. 13, p. 100190, 2025, doi: 10.1016/j.nlp.2025.100190. (Q1)
- M. M. Islam, A. Das, K. Shams, M. R. Hasan, K. F. Hasan, M. R. Rashid, A. Chowdhury, M. S. Ali, M. Islam, M. Shahjalal, and S. Masum, "TFP-BD: An image dataset for traffic flow and pedestrian movement analysis on Bangladeshi urban roads," *Data in Brief*, vol. 59, p. 111398, 2025, doi: 10.1016/j.dib.2025.111398. (Q1)

5. M. R. A. Rashid, K. F. Hasan, M. R. Hasan, A. Das, M. Sultana, and M. Hasan, "A comprehensive dataset for sentiment and emotion classification from Bangladesh e-commerce reviews," *Data in Brief*, vol. 53, p. 110052, 2024, doi: 10.1016/j.dib.2024.110052. (Q1)
6. M. M. Islam, M. J. Ahmed, M. B. Shafi, A. Das, M. R. Hasan, A. A. Rafi, M. R. Rashid, N. T. Niloy, M. S. Ali, A. Chowdhury, and A. A. Rasel, "BDMANGO: An image dataset for identifying the variety of mango based on the mango leaves," *Data in Brief*, vol. 58, p. 111241, 2025, doi: 10.1016/j.dib.2024.111241. (Q1)

PUBLICATIONS: CONFERENCE PAPERS

1. A. Das, K. Shams, M. R. Hasan, M. R. A. Rashid, R. A. Tuhin, M. S. H. Khan, and R. U. Islam, "Towards annotation-efficient kidney CT scan classification: Supervised and semi-supervised Swin Transformer frameworks," in *Proc. 2025 IEEE Int. Conf. on Signal Processing, Information, Communication and Systems (SPICSCON)*, 2025, pp. 766–771, doi: 10.1109/SPICSCON69221.2025.11504242.
2. S. Hossen, A. Das, M. R. Hasan, and M. R. Huq, "Fast, fault-tolerant, and affordable: A modern data lakehouse ETL pipeline for emerging economies," in *Proc. 2025 28th Int. Conf. on Computer and Information Technology (ICCIT)*, 2025, pp. 3440–3445, doi: 10.1109/ICCIT68739.2025.11491349.
3. A. Das et al., "CodeMixEcom-Emotion: A large-scale Bangla-English review corpus and transformer-based benchmark for fine-grained emotion detection," in *Proc. 5th Int. Conf. on Applied Intelligence and Informatics (AII)*, Washington, D.C., USA, 2025. (Best Paper Award)
4. K. Shams, A. Das, M. R. Hasan, M. Sultana, M. A. Morshed, and A. W. Reza, "Histopathology images-based deep learning prediction of prognosis and therapeutic response in small cell lung cancer," in *Data Mining and Information Security: Proceedings of ICDMIS 2024*, vol. 5, Springer Nature, 2025, p. 283.
5. K. F. Hasan, A. Das, J. F. Salma, R. A. Tuhin, N. Hoque, A. W. Reza, and M. S. Arefin, "Load balancing algorithm: A Markov chain model using queuing theory approach," in *Proc. 2024 Int. Conf. on Innovations in Science, Engineering and Technology (ICISSET)*, 2024, pp. 1–6, doi: 10.1109/ICISSET62123.2024.11003920.

AWARDS AND HONORS

Best Paper Award, 5th International Conference on Applied Intelligence and Informatics (AII 2025), Washington, D.C., USA

Awarded for the paper "CodeMixEcom-Emotion: A Large-Scale Bangla-English Review Corpus and Transformer-Based Benchmark for Fine-Grained Emotion Detection."

RESEARCH AND TECHNICAL PROJECTS

Self-Supervised Malware Classification

- Worked on self-supervised deep learning approaches for malware classification using image-based representations of malware binaries.
- Explored contrastive and masked-image modelling approaches, including SimCLR, MoCo, BYOL, and masked autoencoders, with ResNet-based architectures for label-efficient malware representation learning.

Flower Growth Stage Classification

- Developed a flower growth-stage prediction framework combining four self-supervised learning (SSL) models with a Graph Convolutional Network (GCN).
- Created a custom dataset of eight flower species captured over three months across morning, afternoon, and night conditions.
- Conducted extensive ablation studies on unlabeled data to evaluate representation quality.

Real-Time Safety Kit Detection

- Built a YOLO-based real-time safety compliance monitoring system integrated with Hikvision CCTV cameras.
- Designed the complete deployment setup guideline and automated Telegram alerts with annotated images, triggered only when workers were detected without required safety gear.

Vehicle Detection and Traffic Flow Prediction

- Collected real-world traffic data from multiple junctions across Dhaka and applied a YOLO-based model for vehicle detection.

- Used extracted vehicle information to predict traffic flow patterns through a self-supervised learning approach.

Bangladeshi Product Recommendation Chatbot (Capstone Project)

- Developed an AI-powered product recommendation chatbot to improve the online shopping experience.
- Fine-tuned the LLaMA-2-7B model for query understanding and response generation.
- Used Pinecone vector database for efficient storage and retrieval of product data.

Online E-Ticket System

- Built a web application with HTML, CSS, and JavaScript on the frontend and Django on the backend.
- Implemented seat selection and automated PDF ticket generation.

TECHNICAL SKILLS

Programming Languages: Python, C/C++, SQL, XML, HTML, CSS, JavaScript

Machine Learning and AI: Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Graph Neural Networks (GNN), Large Language Models (LLM), Prompt Engineering

Frameworks and Libraries: PyTorch, TensorFlow, LangChain, Django, FastAPI, NumPy, Pandas

Data and Infrastructure: RAG Pipelines, Vector Databases, MongoDB, SQL Databases, Data Annotation, Data Science

Tools: Git, GitHub, MS Office

PROFESSIONAL DEVELOPMENT

Workshop on Cutting-Edge Technology: Opportunities and Future Skillsets

Bangladesh Computer Council (BCC), BKIICT. Day-long workshop participant.

LANGUAGES

Bangla, English

REFERENCES

Dr. Mohammad Rifat Ahmmad Rashid

Associate Professor, Department of Computer Science and Engineering
East West University, Dhaka, Bangladesh | rifat.rashid@ewubd.edu

Dr. Maheen Islam

Chairperson and Associate Professor, Department of Computer Science and Engineering
East West University, Dhaka, Bangladesh | maheen@ewubd.edu